

VGN Fahrplan & Tickets

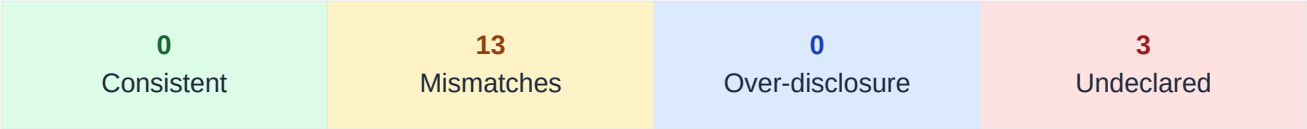
com.mdv.VGNCompanion

Developer	Verkehrsverbund Großraum Nürnberg GmbH
Genre	MAPS_AND_NAVIGATION
Installs	500,000+
Audit Date	2026-04-26

Overall Risk Score

0.76

HIGH



Research prototype · Ongoing research | Policy: <http://www.vgn.de/app/impressum>

Executive Summary

VGN Fahrplan & Tickets declares 13 data type(s) collected and 0 shared in its Play Data Safety label, across 16 data type(s) analyzed. Our heterogeneous GNN audit model identifies **3 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **13 policy-vs-label mismatch(es)**. Overall risk score: **0.76 (HIGH)**.

Top discrepancies by risk:

- **installed_apps**: Undeclared Collection (confidence 0.99)
- **advertising_id**: Undeclared Collection (confidence 0.98)
- **approximate_location**: Undeclared Collection (confidence 0.90)

Class definitions:

CONSISTENT: Label, policy, and SDK signals agree.

POLICY_LABEL_MISMATCH: Label discloses data not mentioned in policy.

OVER_DISCLOSURE: Policy mentions data not declared in label.

UNDECLARED_COLLECTION: SDK signals collection undisclosed in label and policy.

Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

Data Type	Class	Conf.	Evidence
installed_apps	Undeclared Collection	0.99	label:N policy:N sdk:Y (1, 2, 3 +5)
advertising_id	Undeclared Collection	0.98	label:N policy:N sdk:Y (1, 2, 3 +5)
approximate_location	Undeclared Collection	0.90	label:N policy:N sdk:Y (1, 2, 3 +5)
device_id	Policy Label Mismatch	1.00	label:Y policy:N sdk:Y (1, 2, 3 +5)
app_interactions	Policy Label Mismatch	1.00	label:Y policy:N sdk:N (1, 2, 3 +5)
other_info	Policy Label Mismatch	1.00	label:Y policy:N sdk:N (1, 2, 3 +5)
email_address	Policy Label Mismatch	1.00	label:Y policy:N sdk:Y (1, 2, 3 +5)
crash_logs	Policy Label Mismatch	1.00	label:Y policy:N sdk:Y (1, 2, 3 +5)
name	Policy Label Mismatch	1.00	label:Y policy:N sdk:Y (1, 2, 3 +5)
diagnostics	Policy Label Mismatch	1.00	label:Y policy:N sdk:Y (1, 2, 3 +5)
phone_number	Policy Label Mismatch	1.00	label:Y policy:N sdk:N (1, 2, 3 +5)
in_app_search_history	Policy Label Mismatch	1.00	label:Y policy:N sdk:N (1, 2, 3 +5)
user_ids	Policy Label Mismatch	1.00	label:Y policy:N sdk:N (1, 2, 3 +5)
address	Policy Label Mismatch	1.00	label:Y policy:N sdk:N (1, 2, 3 +5)
purchase_history	Policy Label Mismatch	1.00	label:Y policy:N sdk:N (1, 2, 3 +5)
user_payment_info	Policy Label Mismatch	1.00	label:Y policy:N sdk:N (1, 2, 3 +5)

Evidence Trails

Detailed evidence for UNDECLARED_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

Discrepancy: installed_apps — Undeclared Collection (confidence 0.99)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'installed_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: advertising_id — Undeclared Collection (confidence 0.98)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'advertising_id' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: approximate_location — Undeclared Collection (confidence 0.90)

Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Add 'approximate_location' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

Discrepancy: device_id — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'device_id' collection, aligning it with the Play Data Safety label.

Discrepancy: app_interactions — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'app_interactions' collection, aligning it with the Play Data Safety label.

Discrepancy: other_info — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'other_info' collection, aligning it with the Play Data Safety label.

Discrepancy: email_address — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'email_address' collection, aligning it with the Play Data Safety label.

Discrepancy: crash_logs — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'crash_logs' collection, aligning it with the Play Data Safety label.

Discrepancy: name — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'name' collection, aligning it with the Play Data Safety label.

Discrepancy: diagnostics — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'diagnostics' collection, aligning it with the Play Data Safety label.

Discrepancy: phone_number — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'phone_number' collection, aligning it with the Play Data Safety label.

Discrepancy: in_app_search_history — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'in_app_search_history' collection, aligning it with the Play Data Safety label.

Discrepancy: user_ids — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'user_ids' collection, aligning it with the Play Data Safety label.

Discrepancy: address — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'address' collection, aligning it with the Play Data Safety label.

Discrepancy: purchase_history — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'purchase_history' collection, aligning it with the Play Data Safety label.

Discrepancy: user_payment_info — Policy Label Mismatch (confidence 1.00)

Label declares collection: **Yes**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **No** — 1, 2, 3, 4, 5 (+3 more)

Recommended action: Update the privacy policy to disclose 'user_payment_info' collection, aligning it with the Play Data Safety label.

Methodology

PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md.

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

Disclaimer

This report is produced by a research prototype and is provided for informational and research purposes only. It does not constitute legal advice or a regulatory determination. Findings are based on automated weak-supervision labels from publicly available data and have not been reviewed by a privacy legal professional. Research prototype · Ongoing research.